

Design of a Sustainable Community-Based Item Exchange prototype

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Abstract

SwapIt is a sustainable community-based prototype that encourages responsible consumption through item exchange. The prototype allows users to exchange unused items within their communities instead of discarding them. It follows Human-Computer Interaction principles and a User-Centered Design approach to ensure usability and simplicity. The prototype includes features such as item listing, messaging, notifications, ratings, and feedback. Overall, SwapIt supports reuse, community engagement, and sustainable consumption.

Introduction

In recent years, sustainability and responsible consumption have become major concerns due to the increasing amount of waste and overconsumption of resources. According to the World Bank, global waste generation is expected to increase by 70% by 2050 unless urgent action is taken [1]. Many usable items are discarded even though they can still benefit others, creating unnecessary environmental impact and reducing opportunities for reuse. To address this issue, SwapIt was developed as a sustainable community-based design that encourages users to exchange unused items instead of purchasing new products. The prototype was designed to provide a simple, user-friendly, and accessible experience that supports sustainability, community collaboration, and responsible consumption through features such as item exchange, messaging, notifications, ratings, and feedback systems.

Methodology

This study followed a User-Centered Design (UCD) approach to develop and evaluate SwapIt based on user needs and usability principles. Data was collected through an online survey with 92 participants and semi-structured interviews with 18 participants to understand users' attitudes toward sustainability, item swapping, trust, and safety.

The survey results showed strong support for item swapping. Figure 1 shows that 71.7% of participants were willing to continue using item swapping as part of their lifestyle, while Figure 2 shows that 79.3% considered swapping a practical alternative to purchasing new items. However, trust and safety concerns were identified as important challenges.

Based on the collected data and core user tasks, an initial prototype was designed and later refined into a final prototype. The final design included features such as verified profiles, ratings and reviews, search and filtering, favorites, rewards, notifications, and secure in-app communication.

The final prototype was evaluated using heuristic evaluation, discount usability testing, cognitive walkthrough, and remote usability testing. Based on the results, improvements were made to navigation, feedback messages, notifications, trust features, category organization, favorites, and interaction feedback.

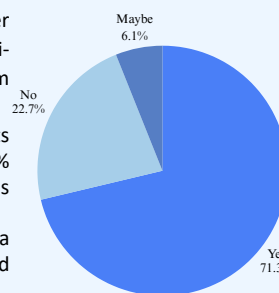


Figure (1)
shows participants' willingness to continue item swapping.

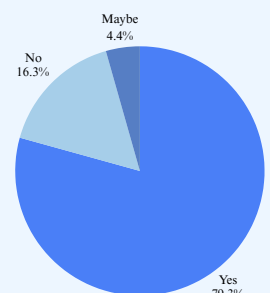


Figure (2)
shows item swapping as a practical alternative to purchasing new items.

Results and Discussion

The final design of SwapIt is the result of an intensive research and testing process focused on enhancing user trust and operational efficiency. Through our study, we identified that privacy and security concerns are the main obstacles for users. To address this, we designed a transparent rating and review interface. When comparing our approach to existing apps like Badalha [26], our design offers a more interactive experience.

A key improvement came from the Usability Testing phase, where we focused on core operations such as managing exchange requests. During the tests, we observed that users experienced uncertainty when transaction statuses were not immediately clear. Consequently, we reached the optimal design by refining the request interface to provide real-time updates such as Pending, Approved, or Rejected status as shown in the Request Management interface Fig. 4. This refinement directly improved the "Visibility of System Status," ensuring a seamless experience.

Additionally, we integrated the "Reward and Badge System" Fig. 3 to boost continuous engagement through gamification. Our testing confirmed that these incentives encourage sustainable sharing behaviors.

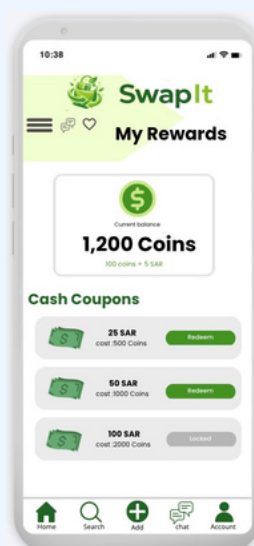


Figure (3)
My Rewards Screen.

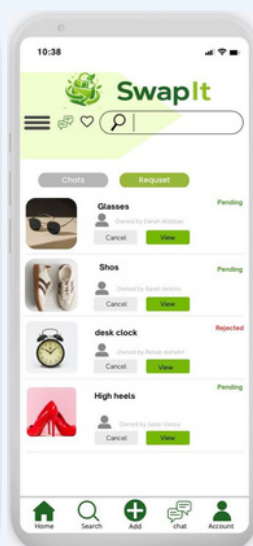


Figure (4)
Request Management Screen.

Conclusion

SwapIt demonstrates how Human-Computer Interaction principles can support the design of sustainable and community-centered digital systems. The prototype encourages responsible consumption, reuse, and community engagement through a simple and user-friendly experience, providing not only a technical solution but also a socially and environmentally meaningful contribution that supports sustainability and reduces waste. In future development, SwapIt can be implemented as a complete mobile application and tested in real-world use to improve user interaction, item exchange, and system performance. Artificial Intelligence can also be integrated to enhance security, detect suspicious activity, improve search accuracy, and provide personalized recommendations and AI-powered user assistance.

References and contact



References



Prototype

